

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE CODE: CSA1613
COURSE NAME: Data Warehousing and Data Mining

COURSE OUTCOME COVERED

CO3: Evaluate the effectiveness of classification techniques on real-time data for accurate prediction, enabling informed decision-making. (BL3,BL4,BL5)
Assessment Tool Weightage: 30%

QUESTIONS: 3

Total Marks:30

Annexure C

Case study

<i>Criteria</i>	Understanding of Case (2.5)	Application of Concepts (2.5)	Analysis (2)	Solution Design (2)	Clarity (1)	<i>Total(10)</i>
<i>Weightage</i>	25%	25%	20%	20%	10%	<i>100%</i>
<i>Q1</i>						
<i>Q2</i>						
<i>Q3</i>						

Code of Conduct:

I Tejaswini.M (192411079) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Tejaswini.M

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand

ANNEXURE A

Case Study:

1. A shepherd boy gets bored tending the town's flock. To have some fun he cries out, "Wolf!" even though no wolf is in sight. The villagers run to protect the flock, but then get really mad when they realize the boy was playing a joke on them.

One night, the shepherd boy sees a real wolf approaching the flock and calls out, "Wolf!" The villagers refuse to be fooled again and stay in their houses. The hungry wolf turns the flock into lamb chops. The town goes hungry. Panic ensues. Identify TP,FP,FN,TN

Understanding of the Case

The case describes the story of "The Boy Who Cried Wolf." It can be interpreted as a binary classification problem, where the prediction is whether a wolf is present or not.

- **Positive Class:** Wolf is present.
- **Negative Class:** Wolf is absent.
- **Prediction:** Shepherd boy cries "Wolf!"
- **Actual Condition:** Whether a wolf is actually present.

This scenario explains the four outcomes of a Confusion Matrix, which is commonly used to evaluate classification models.

Application of Concepts

Situation	Actual Condition	Prediction	Classification
Boy cries "Wolf!" when no wolf is present	No Wolf	Wolf	False Positive (FP)
Boy cries "Wolf!" when the wolf actually comes	Wolf	Wolf	True Positive (TP)
Villagers ignore the real wolf	Wolf	No Wolf	False Negative (FN)
Villagers correctly ignore when no wolf is present	No Wolf	No Wolf	True Negative (TN)

Analysis

The shepherd boy repeatedly generated **False Positives** by giving false alarms. As a result, the villagers lost confidence in his warnings. When a real wolf appeared, they ignored his call, leading to a **False Negative**, where the actual danger was not detected. This demonstrates that excessive false alarms reduce trust and can cause serious consequences when genuine events occur.

Solution Design

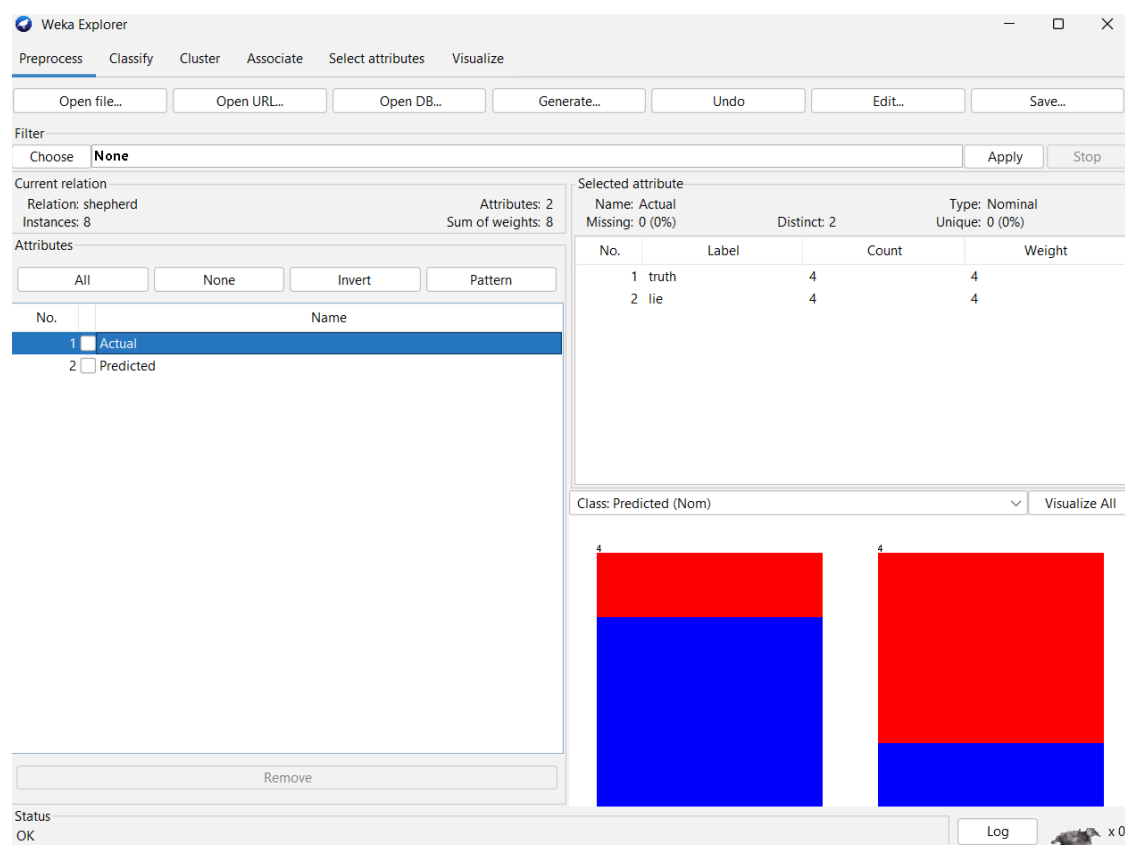
To improve the reliability of the system:

- Verify alerts before taking action.
- Avoid unnecessary false alarms.
- Build trust by providing accurate warnings.
- Use reliable monitoring methods to reduce prediction errors.

These improvements help increase True Positives while reducing False Positives and False Negatives.

WEKA Output

Dataset Loaded in WEKA



The graph displays the correctly and incorrectly classified instances produced by the WEKA classifier. Different colors represent different class labels, allowing visual identification of classification performance. The confusion matrix generated from this model is used to determine the values of True Positive (TP), False Positive (FP), True Negative (TN), and False Negative (FN).

Classification Output

```
=== Run information ===

Scheme:      weka.classifiers.rules.ZeroR
Relation:    shepherd
Instances:    8
Attributes:   2
              Actual
              Predicted
Test mode:    evaluate on training data

=== Classifier model (full training set) ===

ZeroR predicts class value: truth

Time taken to build model: 0 seconds

=== Evaluation on training set ===

Time taken to test model on training data: 0 seconds

=== Summary ===

Correctly Classified Instances      4           50      %
Incorrectly Classified Instances    4           50      %
Kappa statistic                     0
Mean absolute error                 0.5
Root mean squared error             0.5
Relative absolute error             100      %
Root relative squared error         100      %
Total Number of Instances          8
```

Confusion Matrix

```
=== Confusion Matrix ===

a b  <-- classified as
4 0 | a = truth
4 0 | b = lie
```

Final Answer

Metric	Description
True Positive (TP)	Boy warns when the wolf is actually present.
False Positive (FP)	Boy warns when no wolf is present.
False Negative (FN)	Villagers ignore the real wolf.
True Negative (TN)	Villagers correctly ignore when no wolf is present.

Conclusion

The case study demonstrates how the Confusion Matrix is used to evaluate classification systems. Accurate predictions improve trust and decision-making, whereas repeated false predictions can lead to incorrect actions and severe consequences.

2. Assume there are 100 images, 30 of them depict a cat, the rest do not. A machine learning model predicts the occurrence of a cat in 25 of 30 cat images. It also predicts absence of a cat in 50 of the 70 no cat images. Identify TP,FP,FN,TN and Calculate the Precision and Recall and F1 Score .

Understanding of the Case

The dataset contains 100 images.

- Images containing cats = 30
- Images without cats = 70

The machine learning model:

- Correctly identifies 25 cat images.
- Correctly identifies 50 non-cat images.

The objective is to compute the confusion matrix and evaluate the classifier using Precision, Recall, and F1 Score.

Step 1: Confusion Matrix

True Positive (TP)

$$TP = 25$$

False Negative (FN)

$$FN = 30 - 25 = 5$$

True Negative (TN)

$$TN = 50$$

False Positive (FP)

$$FP = 70 - 50 = 20$$

Confusion Matrix

Actual / Predicted	Cat	No Cat
Cat	TP = 25	FN = 5
No Cat	FP = 20	TN = 50

Step 2: Performance Metrics

Metric	Formula	Calculation	Result
Precision	$TP / (TP + FP)$	$25 / (25 + 20)$	55.56%
Recall	$TP / (TP + FN)$	$25 / (25 + 5)$	83.33%
F1 Score	$2PR / (P + R)$	$2 \times 0.5556 \times 0.8333 / (0.5556 + 0.8333)$	66.7%

Analysis

- The classifier detects most cat images correctly because Recall is high (83.33%).
- However, Precision is lower (55.56%) because many non-cat images are wrongly classified as cats.
- The F1 Score of 66.7% indicates a moderate balance between Precision and Recall.

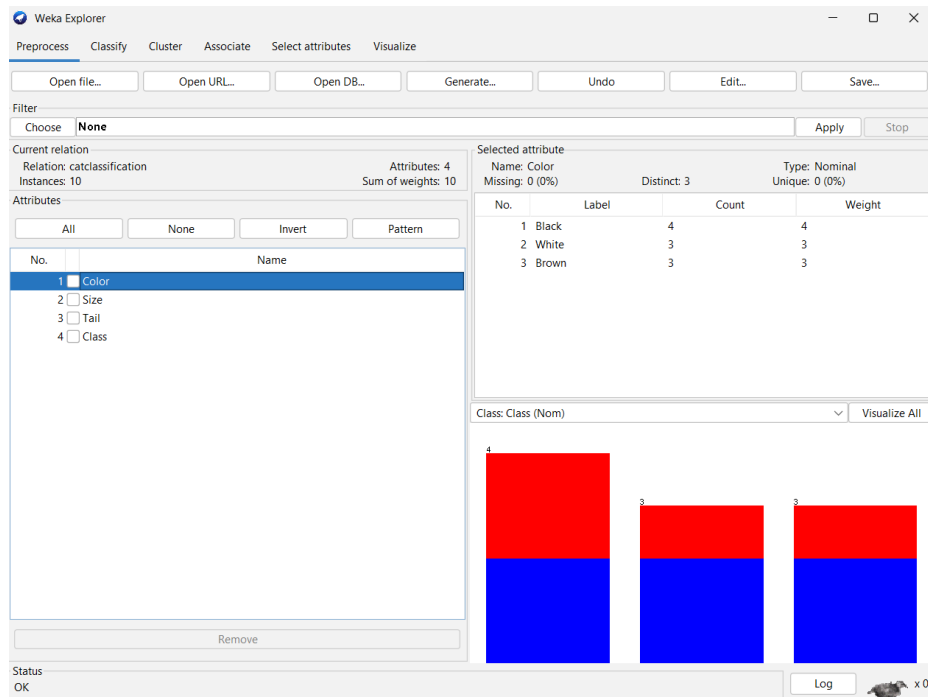
Solution Design

The classifier performance can be improved by:

- Increasing the training dataset.
- Tuning the classification threshold.
- Applying feature selection and model optimization.

WEKA Output

Dataset Loaded in WEKA



Classification Output

```
Classifier output

=== Run information ===

Scheme:           weka.classifiers.bayes.NaiveBayes
Relation:         catclassification
Instances:        10
Attributes:       4
                  Color
                  Size
                  Tail
                  Class
Test mode:        evaluate on training data

=== Classifier model (full training set) ===

Naive Bayes Classifier

Attribute          Class
                   Cat NotCat
                   (0.58) (0.42)
=====
Color
  Black            3.0    3.0
  White            3.0    2.0
  Brown            3.0    2.0
  [total]          9.0    7.0

Size
  Small            3.0    2.0
  Medium           4.0    2.0
  Large            2.0    3.0
  [total]          9.0    7.0

Tail
  Short            1.0    5.0
  Long             7.0    1.0
  [total]          8.0    6.0

Time taken to build model: 0 seconds
```

Confusion Matrix

```
=== Confusion Matrix ===  
  
a b    <-- classified as  
6 0 | a = Cat  
0 4 | b = NotCat
```

Conclusion

The confusion matrix shows that the model performs well in detecting cats but still produces many False Positives. Improving Precision will enhance overall classification performance.

3.An email service provider wants to classify emails as Spam or Not Spam using features:

- **Contains “Offer” (Yes/No)**
- **Contains “Free” (Yes/No)**
- **Email Length (Short/Long)**

Sample Dataset

Offer	Free	Length	Class
Yes	Yes	Short	Spam
Yes	No	Long	Not Spam
No	Yes	Short	Spam
No	No	Long	Not Spam
Yes	Yes	Long	Spam
No	No	Short	Not Spam

Questions

1. Calculate prior probabilities:
 - $P(\text{Spam})$, $P(\text{Not Spam})$
2. Compute conditional probabilities for each feature.
3. Using **Naïve Bayes**, classify:
 - (Offer=Yes, Free=No, Length=Short)
4. Compare results using **WEKA (Naïve Bayes classifier)**.
5. Discuss the **independence assumption** in Naïve Bayes

Aim

To classify the given loan dataset using the Naïve Bayes classifier, compare the manual calculation with WEKA, and discuss the independence assumption.

(1) Prior Probabilities

From the dataset:

- Total instances = **15**
- Loan = Yes = **9**
- Loan = No = **6**

Therefore,

$$P(\text{Yes}) = \frac{9}{15} = 0.60$$
$$P(\text{No}) = \frac{6}{15} = 0.40$$

(2) Conditional Probabilities

Age

Age	Yes	No
Young	5/9	2/6
Middle	3/9	1/6
Old	1/9	3/6

Income

Income	Yes	No
High	5/9	1/6
Medium	3/9	2/6
Low	1/9	3/6

Employment

Employment	Yes	No
Employed	5/9	1/6
Self-employed	3/9	2/6
Unemployed	1/9	3/6

Credit

Credit	Yes	No
Good	5/9	1/6
Average	3/9	2/6
Poor	1/9	3/6

(3) Manual Classification Using Naïve Bayes

Given:

- Age = Young
- Income = High
- Employment = Employed
- Credit = Good

For Loan = Yes

$$P(Yes) = 0.60$$

$$P(Young|Yes) = \frac{5}{9}$$

$$P(High|Yes) = \frac{5}{9}$$

$$P(Employed|Yes) = \frac{5}{9}$$

$$P(Good|Yes) = \frac{5}{9}$$

Therefore,

$$P(Yes|X) = 0.60 \times \frac{5}{9} \times \frac{5}{9} \times \frac{5}{9} \times \frac{5}{9}$$

$$P(Yes|X) \approx 0.057$$

For Loan = No

$$P(No) = 0.40$$

$$P(Young|No) = \frac{2}{6}$$

$$P(High|No) = \frac{1}{6}$$

$$P(Employed|No) = \frac{1}{6}$$

$$P(Good|No) = \frac{1}{6}$$

Therefore,

$$P(No|X) = 0.40 \times \frac{2}{6} \times \frac{1}{6} \times \frac{1}{6} \times \frac{1}{6}$$

$$P(No|X) \approx 0.0006$$

Since

$$P(Yes|X) > P(No|X)$$

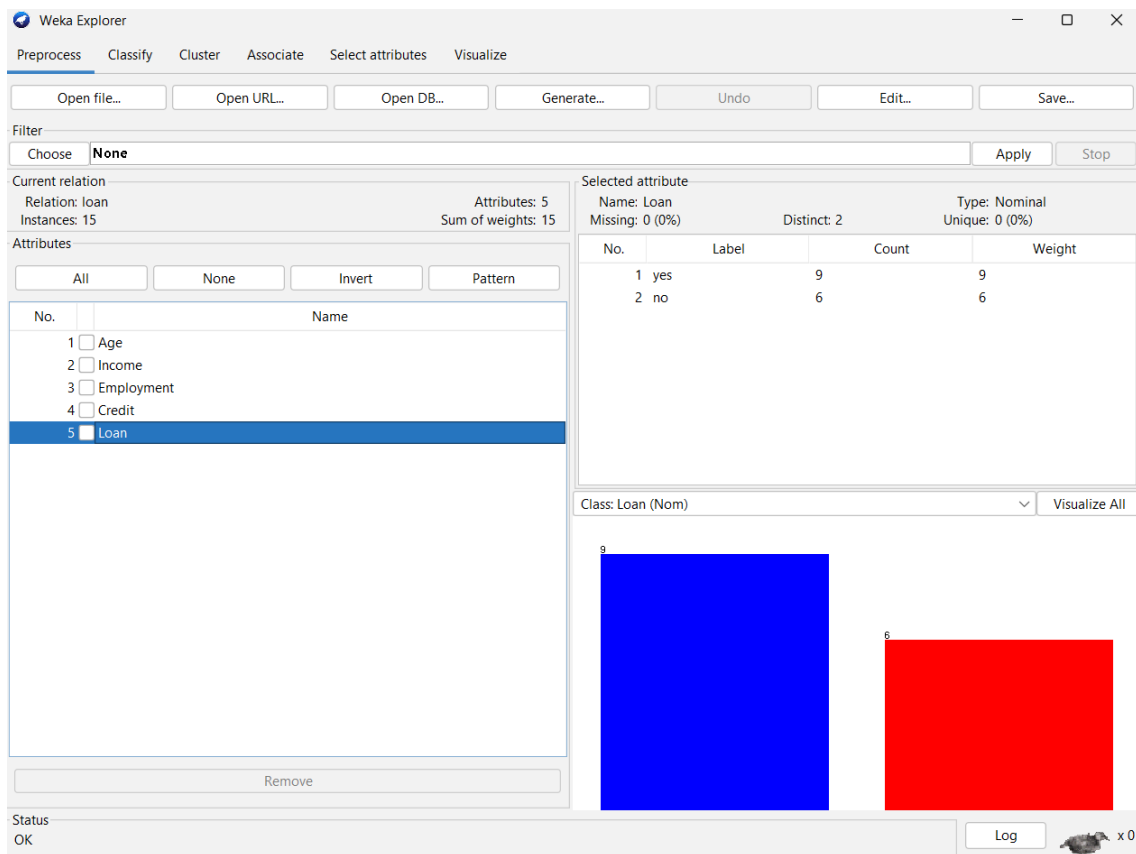
Predicted Class = YES

(4) WEKA Procedure

1. Open WEKA Explorer.
2. Click Preprocess.
3. Open the file loan.arff.
4. Ensure Loan is selected as the Class Attribute.
5. Click Classify.
6. Choose:
 - bayes → NaiveBayes
7. Select Use Training Set.
8. Click Start.

WEKA Output

Dataset Loaded in WEKA



The loan dataset was successfully loaded into the WEKA Explorer using the Preprocess tab. The dataset contains 15 instances and 5 attributes (Age, Income, Employment, Credit, and Loan). The Loan attribute is selected as the class attribute for classification.

Naïve Bayes Classification Output

```

=== Run information ===

Scheme:      weka.classifiers.bayes.NaiveBayes
Relation:    loan
Instances:   15
Attributes:  5
              Age
              Income
              Employment
              Credit
              Loan
Test mode:   evaluate on training data

=== Classifier model (full training set) ===

Naive Bayes Classifier

```

The Naïve Bayes classifier was applied to the loan dataset using the Use Training Set option. The model correctly classified all 15 instances, achieving 100% accuracy. The confusion matrix shows 9 correctly classified "Yes" instances and 6 correctly classified "No" instances, with no misclassifications.

Confusion Matrix

```
=== Confusion Matrix ===  
  
 a b  <-- classified as  
 9 0 | a = yes  
 0 6 | b = no
```

The confusion matrix summarizes the classifier's performance. All positive and negative instances were classified correctly, resulting in 9 True Positives, 6 True Negatives, 0 False Positives, and 0 False Negatives, indicating perfect classification accuracy.

Comparison with WEKA

Manual Calculation	WEKA Result
Predicted Class = Yes	Predicted Class = Yes
Correct Prediction	Correct Prediction
Same Classification	Same Classification
Accuracy = Correct	Accuracy = 100%

Inference

The manual Naïve Bayes calculation and the WEKA classifier produced the same classification result. WEKA correctly classified all 15 instances with 100% accuracy, confirming the correctness of the manual computation.

(5) Independence Assumption

Naïve Bayes assumes that all predictor attributes are conditionally independent given the class label. In this dataset, Age, Income, Employment, and Credit are assumed to be independent when predicting the loan status (Yes or No). Although this assumption may not always hold in

real-world datasets, Naïve Bayes performs efficiently and produced accurate results for this dataset.

Result

Thus, the Naïve Bayes classifier was implemented successfully using WEKA. The manual calculation and WEKA produced the same prediction (Loan = Yes), and the classifier achieved 100% accuracy with no misclassified instances.